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E-201 MEDIUM VOLTAGE SWITCHGEAR

1. SCOPE OF WORK

The Work shall include the design, supply, factory testing, delivery to the Site, installation, testing and commissioning of all medium voltage switchgear as part of the Work in accordance with Part 4, Subsection 1.2 Summary of Work.

The metal clad switchgear described in this Specification shall consist of panels, each complete with the equipment and accessories including CTs, VTs, instrumentation, control, protection devices and interlocking facilities of the system as specified herein and as shown on the Key Single Line Diagram.

The scope of work of each switchgear cubicle shall include, but not be limited to the following:

- (1) Voltage transformer and measurement cubicle
- (2) Incoming breaker cubicles rated as required.
- (3) Outgoing breaker cubicles rated as required.
- (4) Interconnecting breaker cubicles

Spare complete feeder shall be provided at least three (3) feeders for motor and outgoing feeder.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation –Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60028 : International standard of resistance for copper
- IEC 60044 : Instrument Transformers
- IEC 60051 : Direct acting indicating analogue electrical measuring instruments and their accessories
- IEC 60529 : Degrees of protection provided by enclosure (IP Code)
- IEC 62271 : High-voltage switchgear and control gear

2.3 American National Standards Institute

- ANSI C37.20 : Standard for Switchgear Assemblies and Metal-Enclosed Bus.

3. OPERATING AND DESIGN CONDITIONS

3.1 General

All switchgear breakers shall be of either SF₆ or vacuum type designed for indoor service, metal clad design having draw-out type, electrically operated circuit breakers, three-pole, single-throw complying with IEC 62271.

Medium Voltage Auxiliary System Transfer Scheme(s) shall be in accordance with book IV.

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3.2 Equipment Ratings

3.2.1 6.3 kV Switchgear Rating

(1) Circuit Breakers

- a. Number of phases : 3
- b. **Rated Voltage** : 6.3 kV
- c. **Maximum Voltage** : 7.2 kV
- d. Rated insulation level
 - Rated lightning impulse withstand voltage (peak) : 60 kV
 - Rated one (1) minute power frequency withstand voltage (rms) : 20 kV

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- e. Rated frequency : 50 Hz
 - f. Rated normal current
 - Main breakers : *)
 - Tie breakers : *)
 - Feeder breakers : *)
 - g. Rated short-circuit breaking current (rms)
 - : min. 50 kA *)
 - h. Rated transient recovery voltage for terminal faults
 - : 0.345 kV/μs
 - i. Rated short-circuit making current (peak)
 - : min. 125 kA *)
 - j. Rated duration of short-circuit : 1 second
 - k. Rated operating sequence : CO-15s-CO
 - l. Closing mechanism : Stored Energy
 - (2) Bus
 - a. Continuous rating : As required *)
 - b. Momentary rating : As required *)
 - (3) Control bus voltage (nominal) : 220 V DC
 - (4) Stored energy operating mechanism voltage : 220 V DC

Note: *) To be determined by the Contractor based on calculation to the Owner's review and approval

3.2.2 Temperature Ratings

The temperature rise at rated current, above a 40°C ambient temperature of the air surrounding the switchgear, shall not be more than:

- (1) Buses and connections with silver surfaced, welded or brazed joints: 65°C
- (2) Connections to insulated cable (silver surfaced or equivalent) .: 45°C
- (3) Average temperature of the air surrounding outgoing insulated cable: 15°C

- (4) Instrument panels, control panels, doors, cubicles and switchgear enclosures where easily accessible: 30°C
- (5) Parts handled by personnel in the course of normal operation ..: 10°C
- (6) External surfaces inaccessible to operator during normal course of his duties: 70°C

The above temperature ratings shall be in accordance with ANSI C37.20 or equivalent.

The design of circuit breakers shall prevent generation of excessive transient overvoltage caused by breaker operation that may damage the insulation of equipment supplied by the circuit breaker.

Specifically, the transient overvoltage generated by the breaker during the switching operation of typical power plant auxiliary equipment (especially the tripping of stalled small motors and unloaded Power Distribution Center transformers) should be within the voltage magnitude and rise time characteristics given in attached.

4. DETAILED REQUIREMENTS

4.1 General

Medium voltage switchgear shall be manufactured or assembled in Indonesia. The switchgear shall consist of circuit breakers, buses, and accessories mounted on metal frames, complete with all electrical connections, and completely enclosed within sheet metal housings.

4.2 Structures and Housings

The following power equipment shall be fully enclosed in separate sheet steel compartments to provide electrical segregation:

- (1) Each circuit-breaker.
- (2) Each segment of main bus.
- (3) Each set of voltage transformers.
- (4) Each power cable termination complete with current transformers.
- (5) Relays and meters

Each circuit breaker shall be installed in a separate compartment with a

hinged door opening to the front of the housing. The back of each group of vertical compartments shall have a removable cover plate. The metal thickness shall be in accordance with Technical Specification E-280.

Structures shall be designed for removal of the circuit breaker from the front of the housing. A protective device, such as automatically operated shutters, shall be provided.

Groups of switchgear units shall be so designed that future units can be added at either end. Any unused openings shall be covered by removable plates. All faults occurring within any individual cubicle shall be contained within that cubicle and except for busbar faults shall not cause shutdown of any section of the switchgear other than the cubicle itself.

Shutters shall be provided to completely shroud all the fixed isolating contacts of the circuit-breaker buses and feeder circuits. These shutters shall be opened and closed automatically by the movement of the circuit breaker carriage.

It shall not be possible to insert the circuit breaker into the fixed housing portion with the shutters locked in the closed position.

For all switchgear installed indoors in a pressurized and filtered environment and having no fluid pipework or containers, the degree of protection of the enclosure shall be IP 42 in accordance with IEC 60529. For equipment installed indoors in an environment other than as above, the degree of protection shall be IP 54 in accordance with IEC 60529.

Anti-condensation space heaters with thermostatic control shall be installed to provide protection from the effect of moisture and low temperatures.

All mechanical interlocks shall be designed to prevent maloperation at the point where manual power is applied. Padlocking facilities or other approved type of locking devices shall be provided to prevent unauthorized operation or access to the equipment. Locking facilities shall also be provided on cubicle access doors.

All switchgear shall be fitted with an approved telephone jack-in socket which shall be wired to terminals for connecting to the maintenance telephone system described in the Technical Specification E-271, Plant Communication.

The Contractor shall provide and install the switchgear with channel steel bases and mounting bolts.

4.3 Buses and Insulators

Primary buses and connections between the several pieces of apparatus forming the equipment of a panel shall be of high conductivity copper in accordance with IEC 60028 and have a conductivity of not less than 99.9% of that standard. Bolted bus and bar joints and main and auxiliary contacts shall be silver plated.

All buses, connections and their supports shall be of an approved type designed to withstand all normal and abnormal conditions arising in the system. They shall be capable of carrying the short-time current associated with their short-circuit ratings for a period of one (1) second.

Main buses and connections shall be clearly marked to indicate the phase and shall be arranged in the order R, S, T from front to back, top to bottom or left to right as viewed from the switchgear front (operating mechanism side).

4.4 Circuit Breakers

4.4.1 General

Circuit breakers shall be electrically operated, disconnecting type, supported on a frame and designed to be rigidly connected to the housing in the operating position. There shall be provision for easily moving the circuit breaker to a test position (by means of hand-operated device at each breaker), with the main disconnecting devices open, but control and operating circuits closed. The breakers shall be removable, either by separate truck or by wheels located on the circuit breaker.

If the manufacturer's standard arrangement requires a separate truck for insertion and removal of a circuit-breaker from its compartment, then one (1) truck shall be provided for moving each type of breaker in each switchgear room. These trucks shall also be suitable for transferring circuit breakers from one compartment to another.

If the manufacturer's standard arrangement does not require a separate truck for insertion and removal of circuit breakers, then one (1) transfer truck shall be provided for moving each size of breaker in each switchgear room.

Each circuit breaker shall have four (4) positions, i.e. "Service", "Local Test", "Remote Test" and "Disconnected". Marking shall be provided on each position. Provision shall be made to lock the circuit breaker securely in the "Test" and "Disconnected" positions.

Circuit breakers shall be arranged in the compartments so that their non current- carrying metal parts are properly grounded. Manual devices for closing each size of circuit breaker (or charging the closing spring of the circuit breaker) shall be included. Two (2) sets shall be provided for each switchgear room.

Each circuit breaker shall be equipped with the following:

- (1) A test control switch or pushbutton, operable only when the breaker is in the test position.
- (2) A rotary type auxiliary switch with all contacts, whether used or not, connected to the terminal block.
- (3) An operation counter.
- (4) Two (2) time delay under voltage relays, each having two (2) sets of contacts, one (1) wired to the auxiliary relays and the other for audible and remote alarm.
- (5) Sufficient auxiliary contacts and limit switches, etc. for protection, remote monitoring and control purpose.

4.4.2 Operating Mechanism

The circuit breaker operating mechanism shall be motor-charged spring-operated and shall be electrically and mechanically trip-free.

Electrically operated mechanisms shall be designed for remote control at the nominal voltage specified. Closing and tripping mechanisms shall operate satisfactorily over the voltage range indicated below:

- (1) Nominal Voltage : 220 V DC
- (2) Closing Mechanism Operating Range : 158 to 240 (-28% ~ +9%)
- (3) Tripping Mechanism Operating Range : 125 to 245 (-43% ~ +11%)

All circuit breakers shall be provided with stored energy operating mechanisms. It shall be impossible to close a breaker unless there is adequate energy stored to trip the breaker. The breaker springs of electrically operated breakers shall be charged after closing.

Facilities shall be provided to permit manual closing and opening of the circuit breaker in the “Disconnected” position and when the primary contacts are in the open position for maintenance purposes.

Operating Mechanism

- (1) Closing
 - a. It shall not be possible for the circuit breaker to close while the spring is being charged.
 - b. It shall be necessary for the spring to be fully charged before it can be released to close the circuit breaker.
 - c. When the springs are fully charged, the charging mechanism shall be automatically de-actuated.
 - d. A visual mechanical indicating device shall be provided to indicate the state of the spring.
 - e. Facilities shall be provided for manually charging the springs in case of closing supply failure.
 - f. The spring mechanism shall be fitted with local manual release, preferably by push button, shrouded to avoid inadvertent operation.
 - g. There shall be automatic disarming of the charged stored energy mechanism before the circuit breaker is withdrawn from the cell.

(2) Tripping

- a. Each circuit breaker operating mechanism shall be provided with an electrical shunt trip release coil.
- b. Means shall be provided for local manual tripping of the circuit breaker, preferably by push button, shrouded to avoid inadvertent operation.
- c. Facilities shall be provided for locking off the manual tripping of the circuit breaker.
- d. It shall not be possible readily to gain access to the tripping toggle or any part of the mechanism which would permit defeat of the locking of the manual tripping.
- e. It shall not be possible to lock mechanically the trip mechanism so as to render inoperative the electrical tripping.

(3) Method of Control

The method of control shall be arranged as follows:

- a. Remote electrical closing and trip.
- b. Local electrical closing and trip with selection at the circuit breaker.
- c. Local manual closing and trip.
- d. The spring shall recharge automatically after the closing operation has been completed.

4.4.3 Indication of Circuit Breaker Contact Position

A visual mechanical indicating device, readily seen from the front of the circuit breaker unit, shall be provided. It shall be positively driven in both directions so as to show whether the circuit breaker is closed or open when it is in “Service”, “Disconnected” or “Grounded” positions.

4.4.4 Circuit Breaker Auxiliary Switches

Early action auxiliary switches shall be provided on the 6 kV main feed breakers for automatic bus transfer as described in the Specification G-001.

Circuit breakers shall operate as follows:

Position 1, Service Position	:	Primary contacts closed. Remote control operative. Local control non operative.
Position 2, Local Test	:	Primary contacts open. Remote control non operative. Interlocks non operative. Local control operative.
Position 3, Remote Test	:	Primary contacts open. Remote control operative. Interlocks operative. Local control non operative.
Position 4, Disconnected	:	Primary contacts open. Remote control non operative. Local control non operative.

Relay trip circuits shall be operative in each of the above positions except “Disconnected”. Complete interlocking shall be operative in “Remote Test” position regardless of auxiliary switches being mounted on breaker or switchgear. The “Remote Test” shall not be accomplished by plug and receptacle jumpers.

4.4.5 Isolating Features

All circuit breakers shall be connected to their associated buses and cables through isolating devices of an approved design which shall be arranged for operation while the main circuit is live but no current passing.

Isolating devices shall incorporate self-aligning contacts, the fixed contacts of which shall be such that access can readily be obtained for maintenance purposes.

4.4.6 Interlocking and Padlocking Facilities

Mechanical interlocking devices shall be provided to prevent maloperation of circuit breakers.

Padlocking facilities shall be provided for the following:

- (1) Selector mechanisms on circuit breakers; “Disconnected” and “Service” positions.
- (2) Safety shutters on primary contact isolating orifices; closed position only.
- (3) Mechanical trip on circuit breakers; to prevent manual trip

4.5 Grounding

4.5.1 Grounding buses and Connections

Each of switchgear shall be provided with a high conductivity copper grounding bus, which shall run the full length and be bolted to the frame of the switchgear assembly. Grounding connections to the station grounding system and/or building steel shall be provided at each end of the bus. The ground bus dimensions shall in no case be less than 7 x 50 mm copper or equivalent.

4.5.2 Grounding Devices

Cable grounding facilities shall be provided on all incoming and feeder circuits of the switchgear.

A key interlocking system shall be provided so that it is not possible to apply a ground to the buses or cables until all circuit breakers which can feed the bus or cable are open and locked isolated. With the circuit cable or buses grounded, the shutters over the ungrounded fixed main isolating contact shall be closed automatically.

4.6 Terminations

4.6.1 Non Segregated Phase Bus Duct Termination

Bus duct connections shall enter the switchgear vertically from top.

4.6.2 Cable Terminations

Compression type cable lugs for terminating feeder cables entering and leaving the enclosed structures shall be furnished and installed by the Contractor.

4.7 Control Devices and Wiring

Control wiring shall be 2.5 mm² except where large size conductors are needed for current carrying requirements. The minimum conductor size for current transformer secondary circuits shall be 6 mm². The conductors shall be stranded copper for fixed wiring and extra flexible copper for hinge wiring. The conductors shall have 600 volt, 90°C, polyvinylchloride insulation.

Wiring and schematic diagrams covering all electrical equipment that is factory wired as an integral part of this Technical Specification shall be provided.

4.8 Protective Relays and Test Devices

4.8.1 General

The Contractor shall prepare and submit for approval relay coordination curves, both phase and ground fault, for all relays in his supply. The coordination curves shall include relay setting ranges, recommended relay settings and shall be supported by relevant calculations.

Protective relays shall conform to the requirements in the Technical Specification E-250, unless otherwise specified herein.

The protective relays offered shall be static or electro-magnetic of well-proven design but the final output tripping signal shall be from an electrically 'clean' contact.

All protective relays that directly initiate tripping shall be of the hand-reset type. Other protective relays shall be of the self-reset type with contacts to close for sufficient time to ensure the necessary length of tripping impulse.

AC voltage coils on protective relays shall be rated for 220 V at 50 Hz. Flag indicator and holding coils shall be provided on protective relays and shall be suitable for operation between 80% and 110% of rated voltage.

DC auxiliary relays shall have operating coils rated at 220 V DC continuous and shall be suitable for operation between 80% and 110% rated voltage.

4.8.2 Main Feed and Reserve Feed Relays

The following types of relays shall be furnished in each main and reserve circuit breaker for bus faults and backup protection for branch circuit faults:

- (1) Phase overcurrent relay, single element, definite time type, and without instantaneous elements.
- (2) Ground overcurrent relay, single element, definite time type, without instantaneous elements.

Extremely inverse overcurrent relays shall be used in place of the above and be set low enough to provide some transformer overload protection, if the above relays cannot be properly coordinated.

Bus under voltage relays shall be provided in each group of switchgear to protect connected equipment from degraded voltage condition. These relays shall be set to override voltage dip due to motor starts and system faults.

Synchronism-check relays shall be provided to protect against an operator closing a circuit breaker out of synchronism. These relays shall also be suitable of performing the synchronizing duties described under the Technical Specification G-001.

Phase overcurrent relays and ground overcurrent relay shall be provided in the following equipment:

- 6.3 kV Interconnection Buses.

These relays shall be mounted on 6.3 kV switchgear cubicles.

4.8.3 Motor Feeder Relays

The following types of relays shall be furnished in each motor feeder:

- (1) Phase overcurrent relay, single element, long time type and with instantaneous element.
- (2) Ground overcurrent relay, plunger type, short time type, and with instantaneous element.

The ground relay shall be connected to a “donut” type current transformer which encircles all three phases.

4.8.4 Transformer Feeder Relays

The following types of relays shall be furnished in each transformer feeder:

- (1) Phase overcurrent relays, single element, definite time type and with instantaneous elements.
- (2) Ground overcurrent relays, plunger type, short time type and with instantaneous elements.

The ground relay shall be connected to a “donut” type current transformer which encircles all three phases

Extremely inverse overcurrent relays shall be used in place of the above and be set low enough to provide overload protection, if the above relays cannot be properly coordinated.

4.9 Indicating Instruments

All indicating instruments shall comply with IEC 60051.

Each of switchgear shall be equipped with a voltmeter on VT cubicle and a four position meter switch, having positions of “OFF”, “PHASE R-S”, “PHASE S-T” and “PHASE T-R”. Each switchgear cubicle shall be equipped with an ammeter.

4.10 Instrument Transformers

4.10.1 General

Instrument transformers include all current transformers (CTs) and voltage transformers (VTs) used for relaying, metering and checking functions, and shall comply with the requirement of IEC 60044.

All instrument transformers shall be grounded directly to the ground bus, not to the housing.

4.10.2 Current Transformers

Relaying CTs shall be capable of driving the combined burden of the meters, relays and connecting leads up to twenty (20) times the nominal rating without saturating.

Metering CTs shall maintain their rated accuracy with the connected burden for all currents from 10% to 100% of rated nominal CT secondary current.

4.10.3 Voltage Transformers

Each set of voltage transformers and their protective fuses shall be assembled in a separate compartment, and shall be so arranged that the unit can be readily withdrawn from the operating position.

Provision shall be made to prevent personnel from coming in contact with the VT primary fuses until after the primary disconnecting devices are separated by a safe distance and grounded. The secondary shall also be grounded when the VTs are in the draw-out position.

4.11 Nameplates

The nameplates shall be provided for all switchgears. The name plate shall list all switchgear data required by IEC 62271.

Each of switchgear shall be provided with a main label and danger/caution label where applicable. Main labels shall be engraved in both the English and the Indonesian languages.

All components such as switches, fuses, relays and other devices, whether mounted inside or on the surface of the compartments, shall have identifying labels which shall be related to the identifying references on the arrangement drawing, schematic or wiring diagrams. Circuit identification labels shall be provided at the front and rear of all units and an identical label shall be fitted to the front of all withdrawable moving portions. The wording shall be in the English language only.

Danger/caution labels shall have white letters on a red background and when danger or warning labels are fitted they shall not be mounted on removable parts. The size of danger labels shall be such that they are easily readable. The wording shall be in both the English and the Indonesian languages.

4.12 Tools

The following tools shall be provided:

- (1) One (1) circuit breaker removal device provided with wheels and hand operated elevating equipment if required for each switchgear group.
- (2) Four (4) test cabinet with control switch, indicating lights and control cable connecting device.
- (3) One (1) manual operating device per switchgear.
- (4) One (1) grounding device per switchgear.
- (5) Four (4) portable relay test apparatus, with all necessary cables and connectors, for testing the pick up and delays of the ground relay.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor may submit test certificates of tests which have been carried out on identical equipment. The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or rejected.

5.2 Shop Tests

Full tests shall be made by the manufacturer on the complete switchgear equipment in accordance with the tests specified in the relevant IEC recommendations and ANSI C37.20. The tests shall also include the followings:

- (1) Continuity and polarity tests on all coils and circuits.
- (2) Functional tests on all relays.
- (3) Check of electrical and mechanical interlocks.
- (4) Resistance measurement of the operating coils on each breaker.
- (5) Determination of minimum control voltage at which each breaker will trip or close successfully.
- (6) Check successful operation of trip and close mechanism at specified minimum and maximum control voltages.
- (7) Measurement of current at minimum and normal control voltage impressed on each operating coil.

- (8) Determination of speed of opening and closing operations of each size of circuit breaker.
- (9) Tests on current and voltage transformers in accordance with standard production tests covered in IEC 60044 and IEC 60186.

5.3 Site Tests

Certificates and records of all tests shall be submitted to the Owner for approval in accordance with the requirements of Part 4, Subsection 2.6.

After installation on Site all equipment, where applicable, shall be dielectrically tested and shall be operated to prove that the operating gear, tripping devices, protective gear, interlocking and synchronizing are satisfactory and primary and/or secondary injection tests shall be carried out on all protective equipment.

The Contractor shall be responsible for fully commissioning the protective gear, including primary and/or secondary injection tests. This work shall be carried out by a competent engineer who is fully conversant with such tests. The Contractor shall check calibration of all meters, instruments and indicators. All test equipment shall be provided by the Contractor.

The following site tests shall be performed to ensure a satisfactory system. The tests shall include, but not be limited to, the followings:

- (1) Primary injection tests on current transformers to prove ratio and polarity.
- (2) Secondary injection tests to prove relay and meter calibration.
- (3) High voltage tests on buses, including metal enclosed bus, and switchgear.
- (4) Mechanical tests.
- (5) Point-to-point wiring continuity checks on all low voltage circuits to confirm the final connections agree with the panel wiring diagrams.
- (6) Insulation resistance test at the appropriate voltage of all high voltage circuits.
- (7) Energizing of all low voltage circuits.
- (8) Check operation of all meters, relays, instruments, alarms, and trips.

- (9) Adjustments of all relays in line with the settings to be agreed with the Owner.
- (10) Functional testing of all equipment.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements of included in Part 4, Subsection 2.7. These shall include, but not be limited to, the followings:

- (1) Design Criteria and Engineering Calculations
- (2) Material Certification
- (3) Continuity and Wiring Tests
- (4) Short Circuit Testing
- (5) Insulation and High Voltage Tests
- (6) Mechanical Tests
- (7) Functional Testing
- (8) Packaging and Shipping
- (9) Welding
- (10) Cleaning and Painting

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Material Certificates
- (2) Shop Test Reports

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